DiaTwin

Diabetes Digital Twin

A real-time pharmacodynamic platform for insulin-requiring diabetes, turning passive logging into predictive self-care.

IOB Tracking

MOB Modelling

Dose Advisory



8.7/10

Clinical Utility Score

50K+

Meal-outcome pairs validated

Advisory use only — does not replace clinical judgement

THE PROBLEM

Managing T1D is a 24/7 balancing act



The Precision Gap

Doctor-set therapy plans are complex to apply daily. Patients lack tools to translate clinical parameters into real-time decisions.



Data Without Insight

CGM and finger-prick devices show current glucose levels but offer no guidance on how much insulin to take or when.



Risk of Harm

Without IOB/MOB awareness, insulin stacking causes dangerous hypoglycemia. High cognitive burden leads to dosing errors.

WHAT IS DIATWIN?

A pharmacodynamic digital twin for T1D

DiaTwin builds a real-time model of each patient's glucose state tracking how insulin and carbohydrates behave physiologically to answer three critical questions every minute:

- **What is my BG right now?**
- **Where will it go in the next few hours?**
- **Is it safe to eat or dose?**

IOB Insulin On Board

Insulin still pharmacologically active, calculated via trapezoidal integration

MOB Meal On Board

Carbohydrates still being absorbed, modelled via gamma-distribution kinetics

Net Net BG Effect

Meal impact minus insulin impact, positive = rising, negative = falling

Advisory use only — does not replace clinical judgement

FOR PATIENTS

Predictive Self-Care



Real-Time IOB & MOB

See exactly how much insulin and meal carbs are still active, eliminating guesswork before a correction dose.



Advisory Bolus Calculator

Dose suggestions accounting for current BG, active IOB, planned meal, activity level, and time-of-day resistance.



Pre-Bolus Timing

Recommends optimal pre-meal injection window (0–20 min) based on the glycaemic index of the logged meal.



Structured Meal Logging

Extensive food database including local cuisine categories reducing carb-counting burden and improving adherence.



BG Rise Prediction

MOB is converted into an estimated future glucose rise letting patients anticipate hyperglycaemia before it occurs.



Activity Adjustment

Physical activity logging automatically adjusts dose suggestions supporting safer exercise management.

FOR PHYSICIANS

Evidence-Based Remote Monitoring

Remote Patient Monitoring

Authorised physicians access a dedicated dashboard: BG history, meal logs, insulin records, and activity data for informed pre-consultation review.

Glycaemic Response Analysis

Meal-glucose correlation charts reveal postprandial response patterns across days or weeks objective evidence for ICR or ISF titration.

Clinical Parameter Review

All patient-specific constants (ICR, CF, target glucose, absorption modifiers) are visible in a structured view.

Pattern-Based Titration

Time-in-range statistics, SMBG analysis, and glycaemic variability metrics (MAGE, CV%) provide a structured evidence base for dose conversations.

Consent-Gated Access

Bilateral authorisation model ensures patient data sovereignty. Physicians see only consented patients — GDPR-aligned clinical data governance.

Multi-Patient Overview

Full panel of authorised patients, each searchable by name, enabling efficient review workflows in specialist clinics or GP settings.

MARKET

A Growing, Underserved Market

+5M

Insulin-pen users lacking structured dosing guidance

8.7%

CAGR for global diabetes digital management market

~26K

Conservative market penetration within 3 years

€100/yr

DiaTwin cost vs €3–5K/yr for CGM hardware

Competitive Landscape

Feature	DiaTwin	CGM Only	Basic Apps	Insulin Pumps
Intelligent Dosing	✓ IOB, MOB, Projection	✗	✗	✓ Closed-loop only
Works with Insulin Pens	✓	✓	✓	✗
Annual Cost	~€100	€3,000–5,000	Low	€10,000–16,000

ROADMAP

Path to CE IIa Certification & DiGA

Now

TRL 4–5

- Core algorithms validated
- Web + mobile prototypes
- Clinical utility 8.7/10

Phase I

EXIST Funding

- Team expansion
- Full-stack dev

Phase II

TRL 6–7

- Clinical trial N=100, 6 months
- Safety/efficacy validation
- Physician workflow integration

Phase III

Regulatory

- CE Class IIa certification
- DiGA pathway approval
- International expansion

DiaTwin

Diabetes Digital Twin

From reactive monitoring
to predictive self-care.

- Pharmacodynamic precision for every patient, every dose
- Real-time IOB + MOB — glucose predicted, not just measured
- Physician-supervised, patient-centred, GDPR-compliant

Advisory use only — does not replace clinical judgement